

Module 4: Leveraging AutoML Assignment

Course: EAI6020 - AI System Technologies

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1. Dataset Selection & Parameter

Dataset Selection

For this assignment, I selected the E-Commerce Customer Churn dataset available on Kaggle (Verma, 2021). This dataset contains **5,630 records** and **20 features** capturing customer behavioral and demographic attributes, including tenure, order count, satisfaction score, preferred payment method, days since last order, and cashback amount. The target variable is a binary column labeled **Churn**, where a value of 1 indicates the customer has churned and 0 indicates they have not.

This dataset was selected because it represents a real-world problem with a clear prediction target and natural class imbalance. As Saito and Rehmsmeier (2015) demonstrate, class imbalance is precisely the condition under which the precision-recall curve provides a more meaningful evaluation signal than overall accuracy or ROC AUC, making this dataset well-suited for the evaluation objectives of this assignment. The 20 available features span multiple dimensions of customer behavior, including transactional, demographic, and preference-based attributes, giving AutoML a broad range of signals to work with. The goal of using this dataset is to identify customers at risk of churning so that retention efforts, like a personalized offer or a support follow-up, can be triggered before the customer churns.

Parameter Selection

The training parameters were configured as follows: the objective was set to classification with **Churn** as the **target column**, **all feature transformations** were set to **Automatic**, the **region** was set to **us-central1** to align with the Cloud Storage bucket location, and a **budget of 1 node hour** was allocated with early stopping enabled. The **CustomerID column was excluded** prior to training, as it is a unique row identifier with no predictive signal and would introduce noise rather than meaningful structure into the model. The dataset and parameter configuration were well-suited for the goals of this assignment, as the 20 retained features covered a range of customer behavior dimensions and the binary target variable aligned directly with Vertex AI's classification objective. Because customer acquisition costs five to seven times more than retention (Nasir, 2015), a churn model that identifies at-risk customers delivers measurable value by enabling cheaper retention efforts over costly acquisition campaigns.

2. Training

The model was trained using Google Cloud Vertex AI AutoML, a platform that supports tabular classification through configurable training pipelines, including parameter selection, transformation type specification, and optimization objective configuration (Google Cloud, 2026). AutoML automates many parts of the machine learning pipeline, including preprocessing, feature engineering, model search, and hyperparameter tuning (He et al., 2021). This approach was selected because AutoML removes the need for manual model selection and tuning, systematically searching across a broad configuration space to identify the best-performing model for a given dataset and objective.

The training process runs as a pipeline where each step produces output that becomes the input for the next step. This makes each step easier to audit and reproduce. By setting feature transformations to Automatic, Vertex AI determined the appropriate encoding and scaling strategy for each column type without requiring manual specification, applying different treatments to numeric features such as tenure and order count versus categorical features such as preferred payment mode and login device. Early stopping was triggered prior to exhausting the full 1-node-hour budget, indicating that the model stabilized before the budget ran out.

The pipeline executed successfully, progressing from data ingestion and preprocessing through architecture search, model assembly, and evaluation metric generation. The architecture search component evaluated multiple model configurations automatically, which reflects how AutoML frameworks reduce human effort by automating construction and optimization (He et al., 2021). The final model was selected based on validation set performance across multiple candidate architectures. To understand how well it generalizes, I evaluated its precision-recall tradeoff, particularly important given the class imbalance in this dataset.

3. Evaluation

The precision-recall curve was selected as the primary evaluation tool because the dataset exhibits significant class imbalance, a condition under which accuracy and ROC AUC can produce misleadingly optimistic assessments of model performance. Because the negative class dominates this dataset, ROC curves can inflate apparent performance by counting true negatives, whereas precision-recall curves focus specifically on the positive class we want to predict (Saito & Rehmsmeier, 2015). In this case, precision tells me how many of my flagged customers were actually churners, while recall tells me how many actual churners the model caught. A PR AUC of 1.0 means the model held near-perfect precision across every possible threshold, not just the one I selected.

The shape of the precision-recall curve provides further insight into model behavior. A curve that stays close to the top-right corner shows the model maintains near-perfect precision across most recall values, only dropping precision at the highest recall levels. The precision-recall by threshold chart confirms this pattern: both precision and recall

remain near 100% across most confidence thresholds, with precision staying stable and flat from high thresholds down to approximately 0.46, at which point recall first reaches 100% while precision remains at 99.6%. This behavior indicates that the model is not simply defaulting to majority class predictions to achieve high scores, but is genuinely discriminating between churners and non-churners across a wide range of operating points.



Figure 1. Precision-recall curve, ROC curve and Precision-recall by threshold at confidence threshold of 0.46

The Vertex AI evaluation tab produced a PR AUC of 1.0 and an ROC AUC of 1.0, with a log loss of 0.022. At the default confidence threshold of 0.5, the model achieved a micro-average precision of 99.8%, a micro-average recall of 99.8%, and a micro-average F1 score of 0.998.

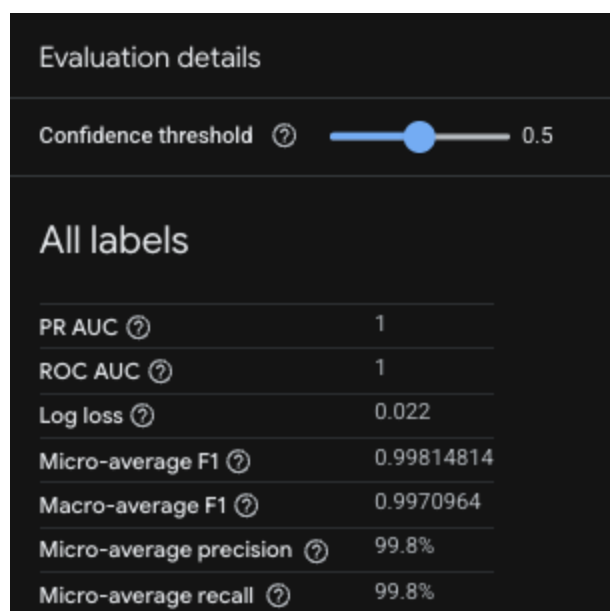


Figure 2. Evaluation Details at confidence threshold of 0.5

At the selected threshold of 0.46, the model achieved a micro-average precision of 99.6%, a micro-average recall of 100%, and a micro-average F1 score of 0.998.

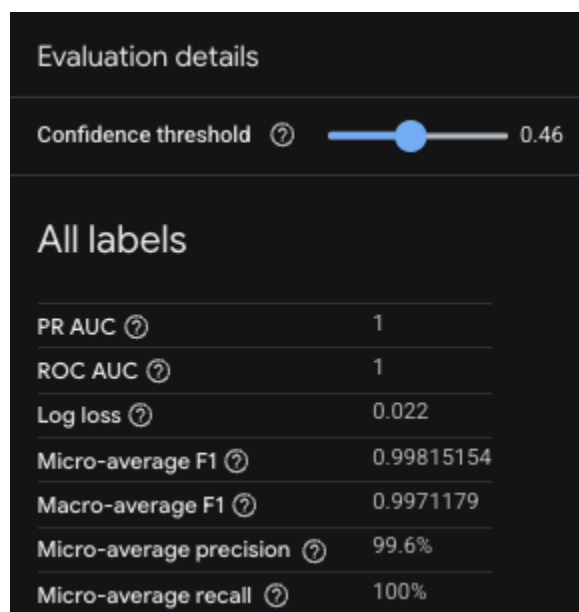


Figure 3. Evaluation Details at confidence threshold of 0.46

The confusion matrix at this threshold showed 432 true negatives, 107 true positives, 1 false negative, and 0 false positives across the evaluation set.

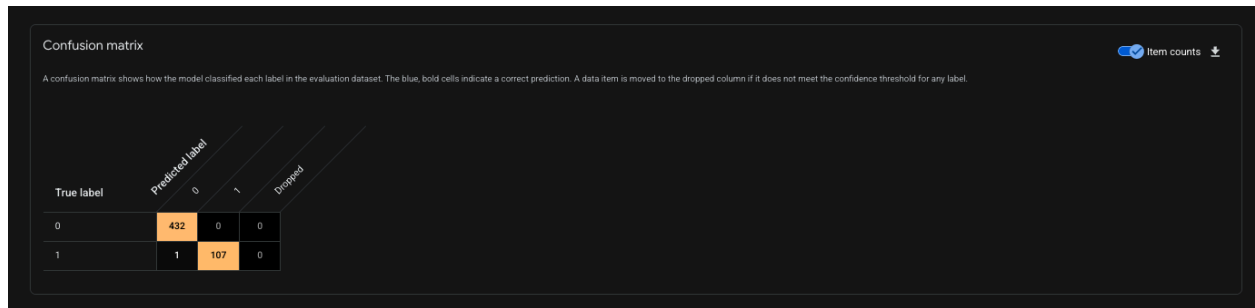


Figure 4. Confusion Matrix at confidence threshold of 0.46

Feature importance attribution, computed using the Sampled Shapley method (Google Cloud, 2026), identified **Tenure** as the strongest predictor at approximately 21%, followed by **Complain** at approximately 11% and **NumberOfAddress** at approximately 8%, indicating that customer longevity and complaint history are the most influential drivers of churn risk in this dataset.

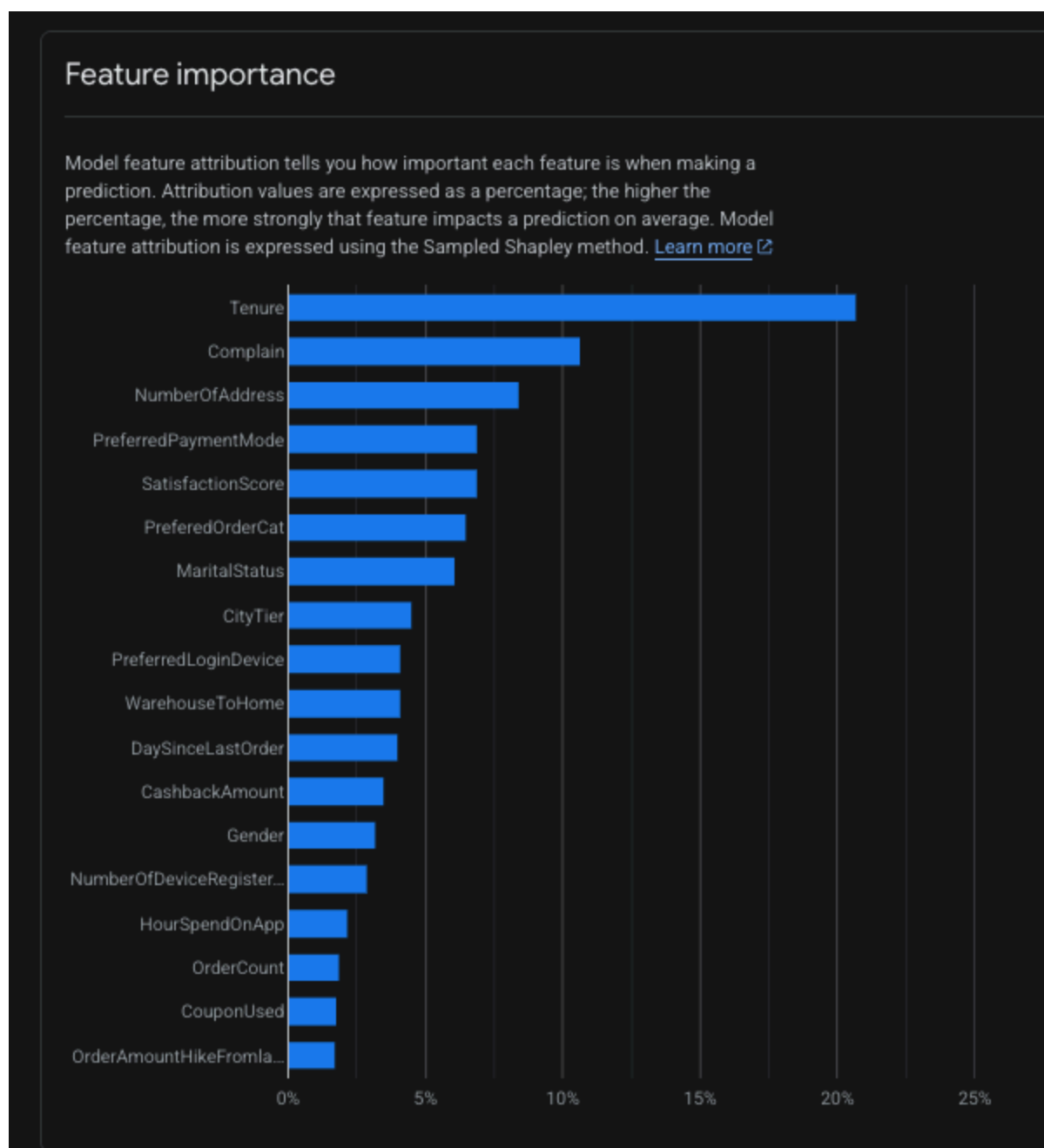


Figure 5. Feature Importance chart at confidence threshold of 0.46

The model achieved a PR AUC of 1.0 and an ROC AUC of 1.0, substantially exceeding the random baseline and confirming the model learned real patterns rather than just predicting the majority class. However, a PR AUC of 1.0 and near-perfect confusion matrix performance warrant careful interpretation. Evaluation metrics computed on a held-out split drawn from the same distribution as the training data may overstate generalization performance, particularly when the dataset is relatively small and the

model has sufficient capacity to fit the underlying patterns tightly. The near-perfect results are consistent with a dataset where a small number of features, particularly Tenure and Complain, carry strong and cleanly separable signal, rather than necessarily indicating overfitting.

4. Score

Choosing a score threshold is ultimately a business decision because it determines how the model balances precision and recall. The default threshold of 0.5 assumes equal cost for false positives and false negatives, which is rarely appropriate in a churn prediction context. In e-commerce, the cost of a false negative, which is failing to identify a customer who subsequently churns, is substantially higher than the cost of a false positive, which results in an unnecessary but low-cost retention offer being extended to a loyal customer. This justifies selecting a threshold below 0.5 to prioritize recall (Nasir, 2015). Saito and Rehmsmeier (2015) recommend that threshold selection should reflect the relative business cost of each error type and the problem's actual cost structure rather than default classifier behavior.

After examining the precision-recall by threshold curve for the positive class, I selected a **confidence threshold of 0.46**. This curve plots how precision and recall for the churning class specifically change as the threshold is adjusted, and it is the correct basis for threshold selection in this context rather than the aggregate micro-average metrics, which combine both classes and can obscure the tradeoff relevant to the business decision. At threshold 0.46, recall for the positive class first reaches 100% while precision remains at 99.6%, making this the optimal operating point on the curve. This threshold was chosen over the default of 0.5 because lowering the bar from 50% to 46% confidence is sufficient to capture the one churning customer the model missed at 0.5, at the cost of only a 0.2 percentage point reduction in precision.

The threshold of 0.46 is also preferable to more aggressive reductions like 0.3. At 0.3, recall stayed at 100% but precision dropped to 98.5%. The curve shows that 0.46 is the point where recall first maxes out before precision starts declining. The selected threshold therefore represents the optimal operating point on the curve, maximizing recall for the positive class without sacrificing meaningful precision, and directly reflecting the business priority of retaining at-risk customers over minimizing retention campaign costs.

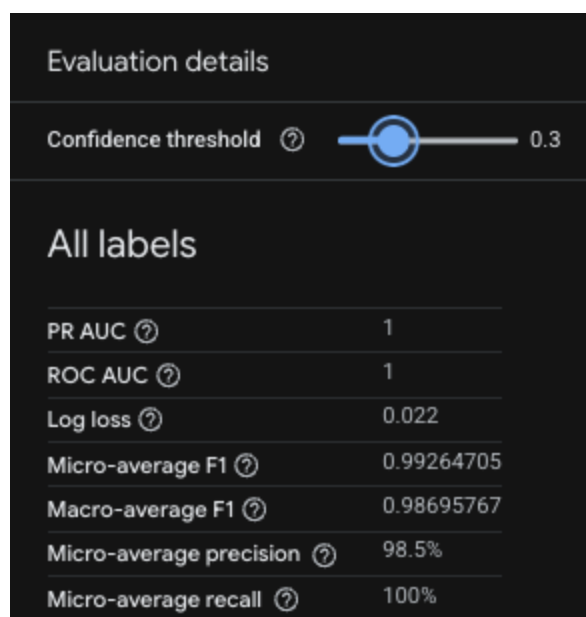


Figure 6. Evaluation Details at confidence threshold of 0.3

5. Lessons Learned

One key lesson I learned from this project was that automated pipelines do not eliminate the need for deliberate feature preparation. Excluding the CustomerID column before training was a necessary preprocessing step that AutoML could not perform autonomously. Had this column been included, the model may have learned spurious correlations between arbitrary identifier values and churn labels, producing inflated training performance that would fail to generalize. This project confirmed what He et al. (2021) found: data quality and feature selection still matter most, even with automated pipelines. In future projects, I will treat feature audit as a mandatory step before any AutoML job is submitted, rather than relying on AutoML to resolve data quality issues independently.

Next, near-perfect evaluation results require critical interpretation, not just celebration. The model achieved a PR AUC of 1.0 and 100% recall at the selected threshold, which initially appears to be an ideal outcome. However, results at or near perfect performance are themselves a signal that warrants scrutiny: they may reflect a dataset where the positive class is highly separable, a small evaluation set that does not fully stress-test generalization, or a combination of both. When results approach perfection, as they did here, the curve provides limited opportunity to observe the tradeoff between precision and recall that makes the plot most diagnostically useful. In future projects, I will treat near-perfect PR AUC as a prompt for additional validation, such as testing on a held-out time period or an independent data source, rather than as a final confirmation of production readiness.

A third lesson I learned was threshold selection should be treated as a collaborative process between technical and business stakeholders. The optimal threshold cannot be determined from the precision-recall curve alone. It requires knowing the relative cost of false positives and false negatives in the specific operational context, including the average revenue lost per churned customer, the cost of a retention campaign, and the capacity of the customer service team to act on flagged predictions. In this project, **the threshold of 0.46 was selected** based on the asymmetric cost structure of churn prediction, where missing a churner is more costly than incorrectly flagging a loyal customer. In future projects, I will present decision-makers with the full precision-recall tradeoff curve rather than a single threshold recommendation, so stakeholders can choose their operating point based on actual costs, not a default setting.

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